

LAB EXPERIMENT RECORD

SET 3: EMPLOYEE PERFORMANCE EVALUATION

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Day	Tuesday

Aim

To visualize employee performance data using a line chart, bar chart, and scatter plot in R, and to build an interactive dashboard using Tableau.

Dataset: Employee Performance

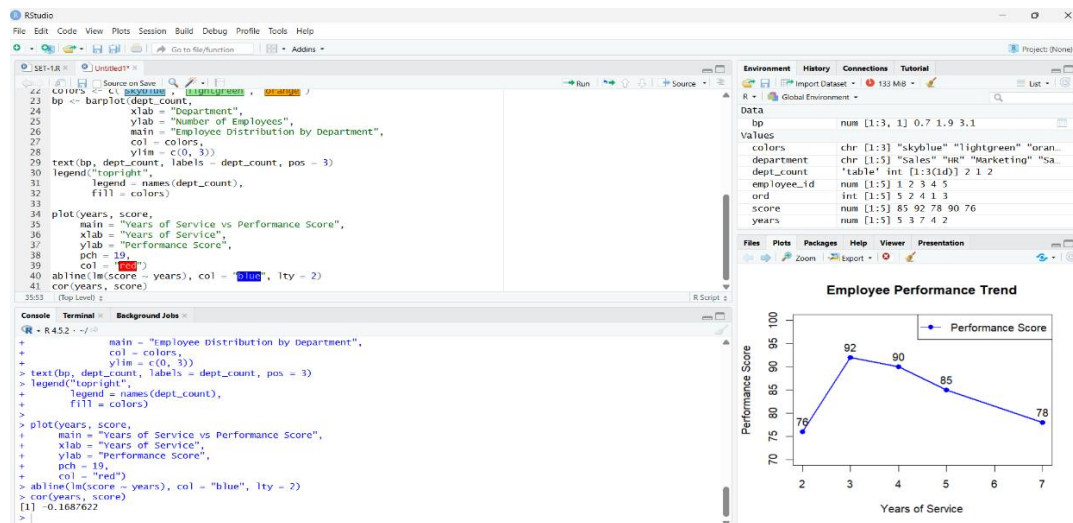
Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

PROGRAM 1: Line Chart of Employee Performance Trend

Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

```
# Data
employee_id <- c(1,2,3,4,5)
department <- c("Sales","HR","Marketing","Sales","HR")
years <- c(5,3,7,4,2)
score <- c(85,92,78,90,76)

# 1. Line Chart
ord <- order(years)
plot(years[ord], score[ord],
      type = "o",
      pch = 19,
      col = "blue",
      lwd = 2,
      ylim = c(70,100),
      xlab = "Years of Service",
      ylab = "Performance Score",
      main = "Employee Performance Trend")
text(years[ord], score[ord], labels = score[ord], pos = 3)
legend("topright",
      legend = "Performance Score",
      col = "blue", lty = 1, pch = 19)
```



Output 1: Employee Performance Trend Line Chart

Observation: Performance is lowest at 2 years of service (76), peaks at 3 years (92), and then declines gradually to 78 at 7 years.

The line chart arranges the employees by years of service to show how performance changes over time. Overall, performance improves early in service and then shows a downward trend as years of service increase.

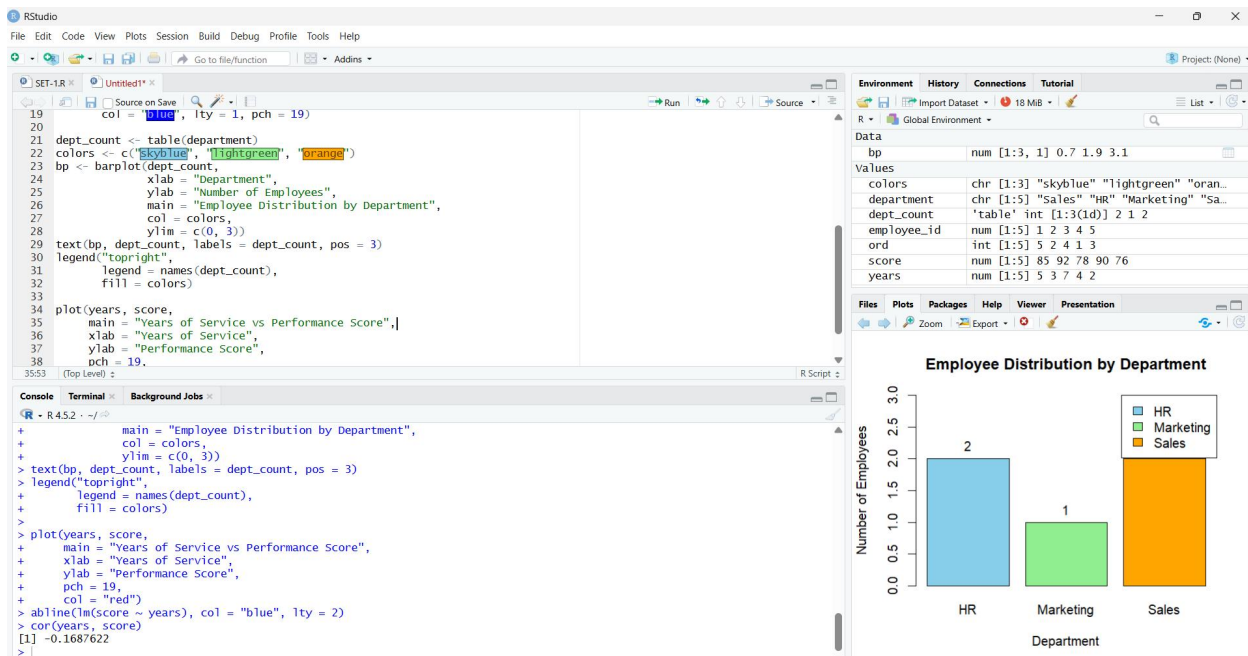
PROGRAM 2: Bar Chart of Employees by Department

Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
# Dataset
department <- c("Sales","HR","Marketing","Sales","HR")

# Bar Chart
dept_count <- table(department)
colors <- c("skyblue","lightgreen","orange")

bp <- barplot(dept_count,
              xlab = "Department",
              ylab = "Number of Employees",
              main = "Employee Distribution by Department",
              col = colors,
              ylim = c(0,3))
text(bp, dept_count, labels = dept_count, pos = 3)
legend("topright",
      legend = names(dept_count),
      fill = colors)
```



Output 2: Employee Distribution Bar Chart

Observation: Sales and HR each have 2 employees, while Marketing has 1 employee.

The bar chart makes it easy to compare the number of employees in each department. Sales and HR have the highest and equal headcount, and Marketing has the lowest with a single employee. The five employees are therefore distributed as 40% Sales, 40% HR and 20% Marketing.

PROGRAM 3: Scatter Plot of Years of Service vs Performance Score

Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

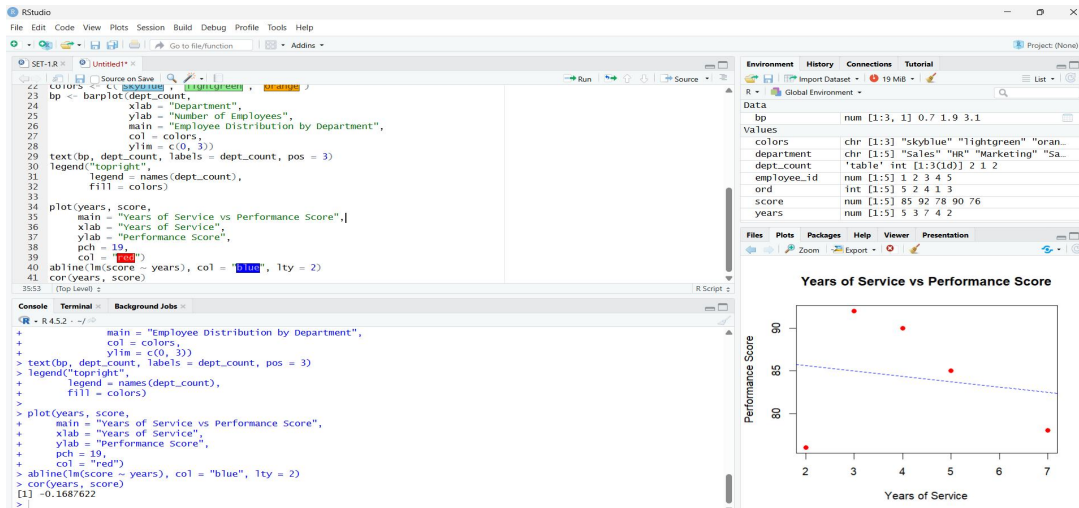
```
# Dataset
years <- c(5,3,7,4,2)
score <- c(85,92,78,90,76)

# Scatter Plot
plot(years, score,
      main = "Years of Service vs Performance Score",
      xlab = "Years of Service",
      ylab = "Performance Score",
      pch = 19,
      col = "red")
abline(lm(score ~ years), col = "blue", lty = 2)

cor(years, score)
```

Console Output:

```
> cor(years, score) [1] -0.1687622
```



Output 3: Years of Service vs Performance Score Scatter Plot

Insight: The correlation coefficient is about -0.17, which indicates a very weak negative relationship between years of service and performance score. The trend line slopes slightly downward, but the points are widely scattered around it. This shows that more years of service do not guarantee a higher performance score; the highest score (92) belongs to an employee with only 3 years of service, while the employee with the longest service (7 years) scored 78.

Note: The dataset has only five employees, so this result is indicative and cannot be treated as a statistically strong conclusion.

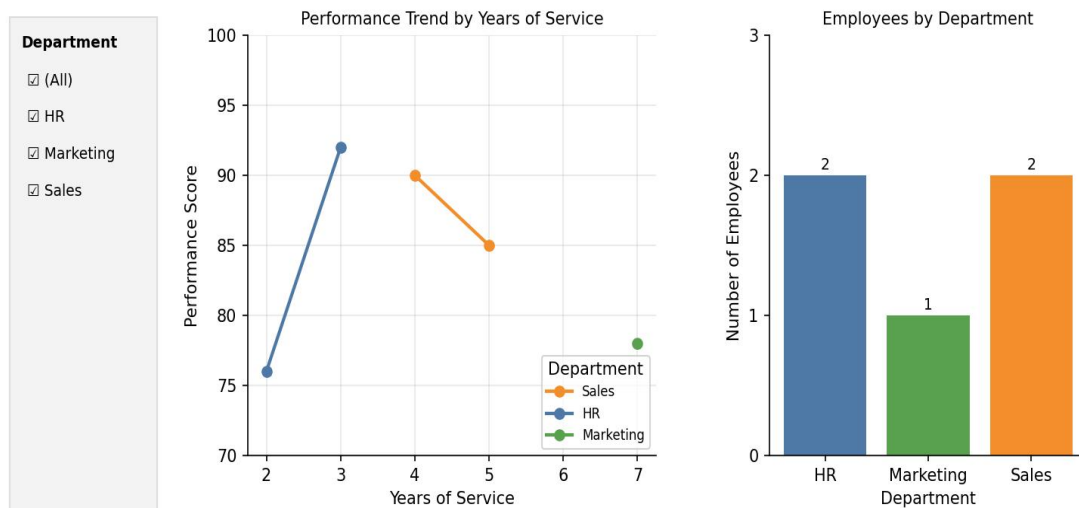
PROGRAM 4: Interactive Employee Performance Dashboard (Tableau)

Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

Tableau Steps:

1. Open Tableau and connect to the data source (Text file / Excel) containing the Employee Performance table.
2. Create Sheet 1 (Line Chart): drag Years of Service to Columns, Performance Score to Rows (SUM), and Department to Color. Add axis titles and show the legend.
3. Create Sheet 2 (Bar Chart): drag Department to Columns and Employee ID to Rows, then change the measure to Count (Distinct). Drag the count to Label to show the values.
4. Open a New Dashboard, set the size to Automatic, and drag Sheet 1 and Sheet 2 onto the canvas.
5. Select the bar chart and enable Use as Filter, then add a Department filter (Show Filter) so that the line chart updates when a department is selected.
6. Give the dashboard the title Employee Performance Dashboard and test the filter and tooltips.

Employee Performance Dashboard



Output 4: Interactive Employee Performance Dashboard

Observation: The dashboard displays the performance line chart and the department bar chart together. Selecting a department in the filter or clicking a bar updates the line chart to show only that department's employees, which allows easy interactive exploration.

Procedure

1. Define the employee ID, department, years of service and performance score vectors in R.
2. Sort the data by years of service and use the plot() function with type = "o" to generate the line chart, then add a legend using legend().

3. Use `table()` to count employees per department and apply `barplot()` to display the distribution with labelled elements.
4. Use `plot()` to draw the scatter plot of years of service against performance score, add a trend line with `abline()`, and compute the correlation using `cor()`.
5. Load the dataset into Tableau, build the line chart and bar chart worksheets, and combine them in a dashboard with filters.
6. Run the dashboard and use the department filter to explore the data interactively.

RESULT:

The employee performance data was successfully visualized using line, bar, and scatter charts in R. An interactive dashboard combining the line chart and bar chart was also created in Tableau for exploring performance by department.

Key Functions and Tools Used

Function / Tool	Purpose
<code>plot()</code>	Creates line and scatter plots.
<code>order()</code>	Sorts data by years of service.
<code>legend()</code>	Adds a legend to the chart.
<code>barplot()</code>	Creates bar charts.
<code>table()</code>	Counts employees in each department.
<code>abline()</code> / <code>lm()</code>	Draws the trend line on the scatter plot.
<code>cor()</code>	Calculates the correlation coefficient.
Tableau Worksheet	Builds the line chart and bar chart.
Tableau Dashboard	Combines charts with filters for interactivity.